

Influence of using wood biomass fly ash as filler on the rheological and physical properties of cementitious pastes

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ABSTRACT

The increasing consumption of biomass by the energy sector is generating a substantial rise in biomass fly ash (BFA). This experimental research explores the use of biomass fly ash as a substitute for limestone filler (LF) in cement-based materials. The study proposes a new recovery pathway for these industrial waste products and addresses the issue of overexploitation of traditional mineral resources. In the research, BFA was first characterized through physico-chemical analyses and compared to limestone filler to identify the similarities between the two materials. Subsequent tests were carried out on ordinary cement CEM I pastes and composite cement CEM III pastes, gradually replacing limestone filler with BFA in volumetric proportions from 0 to 100 %. The results indicate that BFA incorporation reduces flowability, due to increased water demand, and modifies the rheological properties by increasing yield stress and viscosity. In the hardened state, BFA maintains the mechanical properties of the pastes relative to those formulated with limestone filler, while having a minor effect on the physical properties. This study demonstrates that BFA has real potential as a filler in fluid eco-friendly cementitious materials.

List of symbols and abbreviations

α	Degree of hydration (%)
Al_2O_3	Aluminum oxide - alumina
BFA	Biomass Fly Ash
$\text{Ca}(\text{OH})_2$	Calcium hydroxide - portlandite
CaCO_3	Calcium carbonate - calcite
C_2S	Dicalcium Silicate ($2\text{CaO} \cdot \text{SiO}_2$)
C_3S	Tricalcium Silicate ($3\text{CaO} \cdot \text{SiO}_2$)
CSH	Calcium Silicate Hydrate
CEM I	Portland Cement (EN 197-1)
CEM III	Blast Furnace slag cement
DTG	First Derivative of Thermogravimetric analysis
D_{50}	50 % passing particle diameter
D_{90}	90 % passing particle diameter
EDS	Energy Dispersive Spectroscopy

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(continued)

ε_{acc}	Water-accessible porosity (%)
HB	Herschel-Bulkley (rheological model)
K_{HB}	Herschel-Bulkley consistency or viscosity index (Pa.s ⁿ)
$\dot{\gamma}$	Shear rate (1/s).
LF	Limestone Filler
LOI	Loss on Ignition
n_{HB}	Herschel-Bulkley flow index
OPC	Ordinary Portland Cement
SEM	Scanning Electron Microscope
SiO ₂	Silicon dioxide – silica
TGA	Thermogravimetric Analysis
τ_{HB}	Herschel-Bulkley yield stress (Pa)
W/B	Water-to-binder mass ratio
W/C	Water-to-cement mass ratio
XRF	X-Ray Fluorescence
XRD	X-Ray Diffraction

1. Introduction

The biomass energy sector is experiencing significant expansion on a global scale. This dynamic manifests through the increasing number of power plant installations, leading to a significant increase in biomass ash generation [1]. This growth is accompanied by the establishment of numerous power plants, producing more biomass ash. In 2021, the quantity of biomass ash produced by the wood energy sector in France is estimated at 245,000 tons. These considerable volumes of waste are either treated in landfills at high costs or valorized through land spreading, if their chemical properties are suitable [2]. Alongside this waste management challenge, the significant limestone extraction, estimated to exceed 10 million tons per year in France, for the manufacture of cement and limestone filler among other products, renders it one of the most heavily exploited mineral resources [3,4]. This intensive exploitation requires large amounts of fossil energy and affects available resources. Furthermore, there are increasing limitations on the establishment of new quarries.

Limestone filler is a widely used mineral addition in self-compacting concrete. This type of concrete is designed to flow and spread under its own weight without requiring any external energy input [5–8]. A key characteristic of self-compacting concrete is the presence of a significant volume of paste, which can account for up to 40 % of the total concrete volume. The paste plays a fundamental role by enhancing the flowability of the concrete, reducing friction between aggregates, and enabling self-compaction. It typically contains more than 400 kg/m³ of fine particles, coming primarily from cement and mineral additions such as limestone filler [9,10]. By increasing the yield stress and plastic viscosity, the addition of limestone filler reduces sedimentation forces and improves the stability of the self-compacting concrete [11].

To reduce the carbon footprint of concrete, enhance the valorization of industrial waste and decrease reliance on traditional raw materials, numerous studies have investigated the option of substituting cement with biomass fly ash (BFA). Research has demonstrated that the interaction between BFA and the cement matrix are significantly affected by the ash's physicochemical characteristics [12–14]. In fact, the origin of the biomass is a key factor influencing the properties of the ash. Regional industries and locally available sources determine the type of biomass used during combustion [15–17]. Sugarcane bagasse ash in South America and oil palm residues in Asia are often reactive. These ashes typically exhibit pozzolanic activity comparable to coal fly ash and contain significant amounts of amorphous silica, enabling them to be used as cement substitutes at replacement rates ranging from 14 % to 40 % [18–21].

In contrast, wood fly ashes, commonly produced in Europe, are less reactive due to the predominance of crystalline silica, which does not contribute to hydration reaction [22,23]. A decrease in mechanical performance as the substitution rate increases has been highlighted [24,25]. Consequently, several authors have suggested that wood biomass fly ash, because of its inert behavior and fine texture, could be more suitable as a filler material rather than as a reactive binder [26–28]. Using wood BFA as a limestone filler replacement not only reduces the environmental impact of landfilling fly ash but also provides a sustainable alternative to carbon-intensive cementitious materials while conserving natural mineral resources.

Research on biomass fly ash in cementitious materials has predominantly focused on its pozzolanic properties, particularly when used as a partial replacement for cement. Its potential role as an inert filler, especially in the case of wood fly ashes, has received far less attention. A few studies on other biomass fly ash, such as olive residue or sugarcane bagasse, reported enhanced mechanical properties when used as filler in self-compacting or high performance concretes [18,29,30]. However, these findings cannot be directly extended to wood fly ashes due to their distinct chemical and physical nature. In addition, some works have shown that biomass ashes increase the shear stress and viscosity of cement based pastes and mortars [31,32], while some wood ashes may even promote nucleation, enhancing cement hydration [33]. These insights remain partial, and a comprehensive understanding of the role of low reactivity wood BFA functions as a limestone filler substitute is still lacking.

This study addresses this unmet need by providing a detailed experimental investigation into the use of wood BFA as an inert filler in cementitious pastes. Unlike prior studies that mainly focused on the pozzolanic ashes as cement replacements, the present work explores their suitability as filler substitution for limestone. This research thus provides new and practical insights into the valorization of wood BFA as an eco-friendly filler, advancing the understanding of its physical, mechanical, and hydration-related effects, and offering a viable pathway for the sustainable use of this industrial by-product in cementitious materials.

In this study, only fluid pastes were tested in order to better understand the effect of incorporating wood biomass fly ash as a filler.

The paste is a crucial component in self-compacting concrete because it determines the flow behavior and facilitates the movement of aggregates by reducing internal friction [9,10]. Working with pastes allows the effect of BFA to be isolated without the complicating influence of aggregates. BFA replaced limestone filler in different pastes formulations using two binders. The first is CEM I, composed exclusively of clinker, which allowed for the analysis of the substitution effect in a conventional binder matrix. The second is CEM III, a composite cement commonly used in fluid concrete for deep foundations, as recommended by standards EN 1538+A1 and EN 206+A2/CN [34]. In such applications, the cement content must be greater than 325 kg/m^3 , with a quantity of fines greater than 400 kg/m^3 [35]. This composite cement enhances the resistance to chemical attack and reduces the risk of cracking in massive structures. To evaluate the interaction between BFA and a blended binder system, fluid pastes incorporating CEM III were also prepared.

Due to their distinct chemical characteristics and compositions, CEM I and CEM III can interact differently with BFA, thereby imparting varied properties to the pastes. The physicochemical characteristics of the BFA used in the study were analyzed in the first instance to understand its potential interactions with the cement matrix. Then, an analysis of the effect of BFA at various substitution percentages was conducted to identify the optimal incorporation rate.

2. Experimental program

2.1. Materials

The limestone filler used was a commercially available filler composed of 98 % CaCO_3 . The BFA comes from the Lens plant in France, which is equipped with a dense fluidized bed power plant. Biomass is burned in a suspended sand bed using a gas flow. The wood biomass used comes from round wood, pallets and forest chips. The Lens plant produces 33 GWh of heat for the city and 40 GWh of electricity per year through cogeneration. Combustion operations occur at 850°C . The BFA consists of pulverulent ashes recovered after flue gas treatment using bag filters. The plant generates more than 600 tons of BFA per year and 1800–1900 tons of bottom ash [36]. A CEM I 42.5 R composed of 95.5 % clinker and a CEM III/A 52.5 L cement composed of 36 % clinker, 62 % blast furnace slag and 2 % limestone were used for the various pastes.

2.1.1. Chemical and mineralogical compositions

Several parameters influence the chemical composition of BFA, including the technology, combustion temperature and the type of biomass used [12,37,38]. The oxides containing high levels of the biomass fly ash from the study and the loss on ignition (LOI) were identified through chemical analysis using X-ray Fluorescence (XRF). The chemical composition of the BFA is presented in Table 1.

Traces of ZrO_2 , Cr_2O_3 , TiO_2 , ZnO , and Mn_2O_3 are found at less than 1 %. The chemical composition of the wood biomass used and the combustion technology explain the predominance of silica, lime and potassium oxides [12,14,24]. The chemical composition of the biomass fly ash studied reveals a notable similarity with wood biomass ash data reported in the literature. Silica and lime are the majority oxides observed in wood BFA, followed by alumina, potassium oxide and iron oxide. Their levels in the ashes studied are within the ranges typical of some wood residues [2,12,15,24,39]. The presence of unburnt particles (LOI = 15.93 %) has undesirable effects on the fluidity and mechanical strength of cementitious materials, as they absorb large quantities of admixtures and water [24, 40]. According to standard EN 450-1 on the physical and chemical properties of siliceous coal fly ash, it can be considered pozzolanic if the sum of oxides % SiO_2 + % Al_2O_3 + % Fe_2O_3 is greater than 70 %. The oxides also need to be present in amorphous form. In the case of the BFA studied, the sum is 43.84 %.

An X-ray diffractometer (XRD) analysis, shown in Fig. 1, was carried out to identify the presence or absence of an amorphous phase.

The absence of an amorphous phase is observed, corroborating the lack of pozzolanic reactivity. The X-ray diffractogram shows crystalline peaks of mainly quartz (SiO_2) and calcite (CaCO_3). These phases are due to the chemical composition of the biomass and to the combustion process using fluidized bed technology in the presence of silica sand [2,41]. These observations concur with those of several studies highlighting the dominance of crystalline phases and even the absence of amorphous phases in wood BFA [40,42]. A higher abundance of quartz, calcite, lime and portlandite is also noted [2,16,43].

2.1.2. Physical properties

The specific surface area of a mineral addition can influence water demand and, consequently, the properties of cementitious materials in the fresh state [34,44]. To ensure a fair comparison of behavior within the cementitious matrix, ashes were selected with a Blaine fineness similar to that of limestone filler. The specific surface area of the materials was determined using a Blaine permeameter, in accordance with EN 196-6. The Blaine specific surface area of the limestone filler is $4889 \pm 197 \text{ cm}^2/\text{g}$. BFA with a particle size of $200 \mu\text{m}$ achieved a Blaine fineness close to that of filler, with a surface area of $5138 \pm 147 \text{ cm}^2/\text{g}$. The particle size distributions of BFA and limestone filler are presented in Fig. 2. The particle size distribution of the limestone filler shows particles ranging from 0.1 to $100 \mu\text{m}$, with the majority having a median diameter D_{50} of $10.9 \mu\text{m}$ and a D_{90} of $43.6 \mu\text{m}$. This indicates a relatively fine and homogeneous particle distribution. In comparison, the BFA particles have a broader size range from 0.1 to $200 \mu\text{m}$ with a median diameter D_{50} of

Table 1
Chemical composition of the BFA used in mass %.

Oxides	SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	Na_2O	K_2O	P_2O_5	SO_3	LOI	Others
Content (%)	38.03	3.77	2.04	24.65	2.95	1.07	5.31	1.96	2.92	15.93	1.37

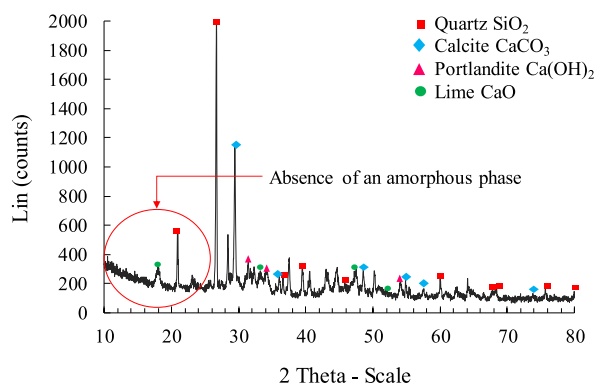


Fig. 1. Mineralogical composition of BFA.

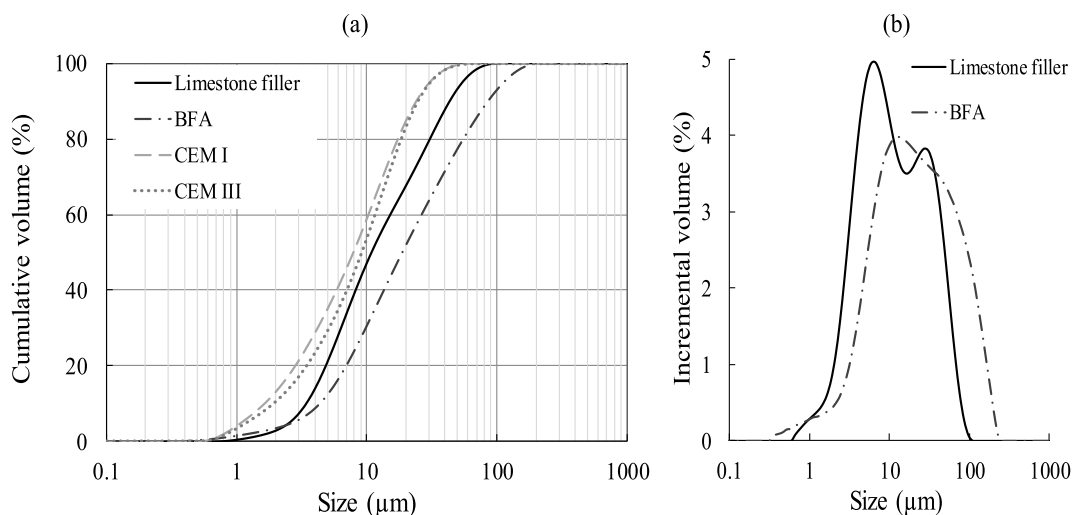


Fig. 2. Particle size distribution of cement, BFA and limestone filler (a) cumulative volume and (b) incremental volume.

18.8 μm and a D_{90} of 85.7 μm . The similar specific surface area found with the coarser BFA can be attributed to the morphology and surface characteristics of the ash particles [45]. This difference in particle size and the slight difference between the specific surface areas of BFA and limestone filler may lead to reduced packing density, potentially increasing void content and the water demand, and thereby reducing flowability [23,46]. CEM I and CEM III have specific surface areas of 4300 cm^2/g and 4150 cm^2/g , respectively.

The density of BFA was determined using the NF P15-435 standard, using ethanol because BFA is likely to react with water. The

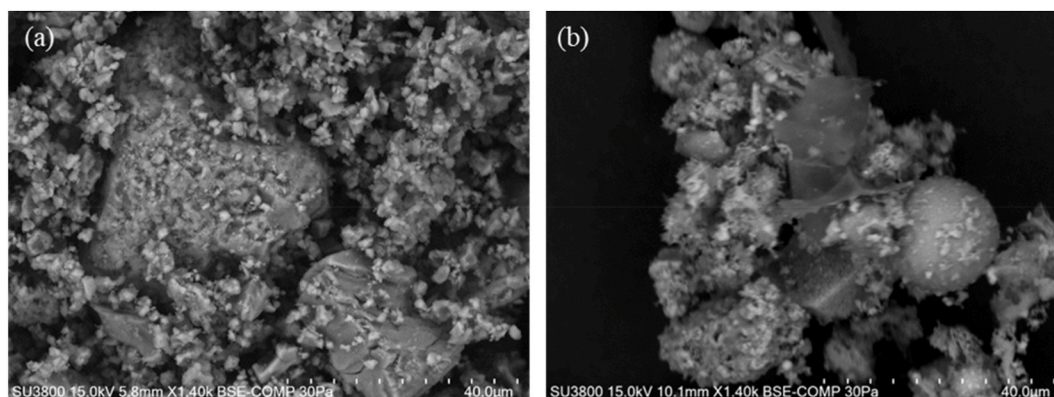


Fig. 3. SEM observation of several grains of (a) limestone filler and (b) BFA.

absolute density of BFA is $2.66 \pm 0.03 \text{ g/cm}^3$, while that of the limestone filler is 2.70 g/cm^3 . The density of CEM I is 3.10 g/cm^3 and that of CEM III is 3.00 g/cm^3 .

2.1.3. SEM observation

Analyses performed using a scanning electron microscope and energy-dispersive X-ray spectroscopy (SEM/EDS) identified the surface characteristics of the biomass fly ash and the limestone filler. The observed particles are shown in Fig. 3. The examination of the limestone filler reveals angular particles with either smooth or rough textures. Additionally, several agglomerates of small, smooth particles are observed. This structure is attributed to the origin of the limestone filler, which results from the grinding of limestone rock [45,47,48]. An analysis of the BFA identified three distinct particle shapes. These include spherical particles, elongated particles and irregular particles that resemble agglomerates of small particles. The elongated particles may correspond to unburnt particles. These observations are consistent with the literature, which describes BFA particles as having a complex shape, diverse morphology, and a porous structure [2,46,49]. The varied morphology of BFA particles, combined with their porosity, contributes to a larger specific surface area, which may impact the fresh-state properties of cementitious mixes.

2.2. Cement paste preparation

The lack of pozzolanic reactivity in the BFA, combined with the presence of fine particles identified during physicochemical analyses, confirms their potential for use as filler in this study. Pastes were formulated with a water-to-binder ratio (W/B) of 0.41 to achieve sufficient fluidity for rheological measurements [50]. Volumetric substitutions of 30 %, 50 %, 70 % and 100 % of limestone filler with BFA were performed, and the compositions are detailed in Tables 2 and 3. The intermediate values (30 %, 50 % and 70 %) are used to identify potential optimal substitution levels, while 100 % substitution assesses the feasibility of complete replacement.

The mixing procedure was based on a previous study analyzing the rheological properties of cementitious materials and was applied identically to both CEM I and CEM III pastes to ensure reproducibility [50]. Dry materials were introduced into the mixer and homogenized for 60 s at low speed. Water was then added, and the mixture was mixed for 45 s. The paste was manually homogenized for 90 s while scraping the sides and bottom of the mixing bowl. This was followed by mixing at high speed for 60 s. After that, manual homogenization was performed for 60 s, followed by a final mix at high speed for 120 s. All samples were cured under identical conditions at 20°C and relative humidity above 70 % until testing.

3. Influence of BFA on paste properties

This analysis aims to identify the optimal substitution threshold that delivers a balance between flowability and viscosity, without compromising the hydration and strength of the pastes.

3.1. Flowability

To assess the influence of BFA properties on fresh pastes, an analysis of flowability is necessary. As there is no standardized test for measuring the flowability of fluid pastes, spread and slump measurements are carried out using a mini CEM (Concrete-Equivalent-Mortar) cone with dimensions $d = 50 \text{ mm}$, $D = 100 \text{ mm}$ et $H = 150 \text{ mm}$. The test is performed by filling the mold in two layers and homogenizing the mixture. The mold is lifted vertically. A slump is taken at the center of the paste, and the average spread diameter is determined by taking two measurements corresponding to two perpendicular diameters.

The results in Fig. 4 show a decrease in flowability for all substitution rates. Increasing the BFA content significantly reduces both the spread and the slump of the mixtures. The observed reduction in flowability can be attributed to several physical characteristics of BFA. As illustrated in Fig. 2, BFA exhibits a coarser particles size compared to limestone filler. Their incorporation can reduce the packing density of the mixture, leading to a reduction in the amount of excess water available to lubricate the particles. Furthermore, the irregular shape and higher water demand of BFA, caused by the unburnt carbon present in BFA can reduce the fluidity of the pastes. These findings align with previous studies reporting reduced flowability when incorporating BFA into cement-based materials [15,42,51]. The suggested causes include the fineness of the ash and the presence of unburnt particles, which increase water demand and reduce the water available for fluidity [40]. To maintain good flowability, adjustments in water content or the addition of superplasticizers could be effective. The study by Cuenca et al. [29] showed that an increase in the dosage of superplasticizer in concretes containing olive residue biomass fly ash allowed for flowability equivalent to that of reference self-compacting concrete. Reduced

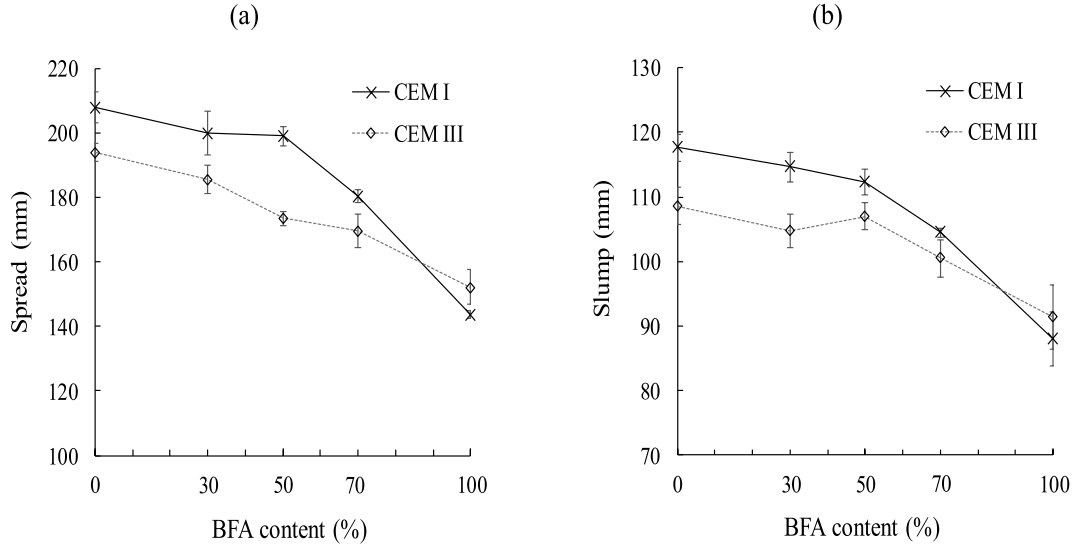
Table 2
Mix proportion of CEM I pastes.

BFA content (%)	0	30	50	70	100
Limestone filler (g)	302	211	151	90	0
BFA (g)	0	89	149	210	297
CEM I cement (g)	1037	1037	1037	1037	1037
Water (g)	554	554	554	554	554
Water/Binder ratio	0.41	0.41	0.41	0.41	0.41
Density (kg/m^3)	1893	1891	1891	1891	1888

Table 3

Mix proportion of CEM III pastes.

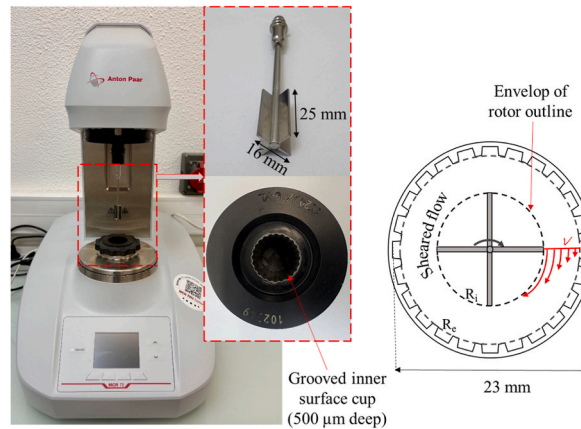
BFA content (%)	0	30	50	70	100
Limestone filler (g)	305	214	153	92	0
BFA (g)	0	90	151	211	301
CEM III cement (g)	1018	1018	1018	1018	1018
Water (g)	548	548	548	548	548
Water/Binder ratio	0.41	0.41	0.41	0.41	0.41
Density (kg/m^3)	1871	1870	1870	1869	1867

**Fig. 4.** Flowability of pastes (a) Spread and (b) slump as a function of substitution rate.

flowability can be managed by using superplasticizers or by keeping the effective water content constant [25,52].

3.2. Rheological properties

The observed reduction in flowability makes it crucial to assess the effect of BFA on the flow of fluid pastes by analyzing their rheological properties. Indeed, rheological properties have a direct impact on the flowability and the stability of the mixtures. Analyzing these properties allows for a better understanding of the decrease in flowability and enables the fluidity of the mix to be adjusted without compromising its stability. Rheological measurements were performed using an Anton Paar MCR 72 rheometer

**Fig. 5.** Rheometer MCR 72 equipped with a four-blade vane geometry and a cylindrical cup milled lengthways. R_i , R_e et v represent the inner cylinder radius, the stationary outer radius and the rotational speed, respectively.

equipped with a four-blade vane geometry (25 mm height, 16 mm diameter). The radial gap between the rotational tool and the external cup was 7 mm. To reduce wall slip effects, the inner wall of the cup has a grooved surface with a depth of 0.5 mm. A schematic representation of the vane geometry is shown in Fig. 5.

A pre-shear of 100 s^{-1} for 60 s was applied to create uniform conditions between each test. After a rest period of 30 s, the shear rate was ramped down from 100 s^{-1} to 0.1 s^{-1} . The shear rate steps and test durations (15 s per step) were set in accordance with a recent study that investigated the rheological properties of cement pastes and compared different rheometers and measurement geometries [50]. The flow curves determined with the rheometer are shown in Fig. 6, illustrating a shear thinning behavior at yield.

These are dense suspensions of colloidal particles, illustrated by the Herschel-Bulkley model.

$$\tau = \tau_{\text{HB}} + K_{\text{HB}} \dot{\gamma}^{n_{\text{HB}}} \quad (1)$$

where τ_{HB} (Pa) represents the Herschel-Bulkley yield stress, K_{HB} ($\text{Pa} \cdot \text{s}^{n_{\text{HB}}}$) the Herschel-Bulkley consistency or viscosity index, n_{HB} the Herschel-Bulkley flow index and $\dot{\gamma}$ is the shear rate ($1/\text{s}$). The rheological properties of the mixtures are summarized in Table 4.

A linear increase in viscosity and a rise in yield stress are observed. As shown in Fig. 7, these increased become more pronounced as the substitution rate increases. The most viscous pastes are obtained, regardless of the cement used, when 100 % of the limestone filler is replaced with BFA. This increase is in accordance with the observations made in the flowability test. This behavior may be linked to the presence of unburnt carbon (15.93 %) which causes BFA particles to demand more water than limestone filler. This water demand increases friction forces and interactions between the fine particles. Chandara et al. [53] have shown that high levels of carbon (>10 %) in ash increase water demand and the necessary superplasticizer dosage.

The effects of BFA on flowability and rheological properties is more pronounced in CEM I pastes. The reduced effect on CEM III pastes may be due to the presence of slag, which preserves better flowability by increasing the fluidity of the cementitious mixture [54, 55]. The replacement of clinker with blast furnace slag in CEM III helps reduce yield stress and viscosity. This is due to the vitrified surface of the slag grains, which enables better sliding between the solid constituents [56].

With increasing BFA content, regardless of the type of cement, the Herschel-Bulkley flow index n_{HB} decreases as shown in Fig. 8a, while the viscosity index K increase (Fig. 7b). These changes reflect a more pronounced shear-thinning behavior, demonstrating BFA's ability to reduce the energy required to overcome interparticle forces. Shear-thinning results from microstructural rearrangements within the mixture. At low shear rates, attractive forces between particles dominate over hydrodynamic forces, leading to the formation of structural aggregates [57]. In shear thinning suspensions, fine particles become organized into layers aligned with the flow direction, minimizing collisions. The presence of some spherical particles in BFA likely aids the formation of flow-aligned layers thereby lubricating the contacts between individual structural grains.

The rheological measurements confirm the findings from the flowability study. As seen in Fig. 8b, a linear correlation exists between yield stress and slump, corroborated by a high coefficient of determination ($R^2 = 0.97$ for CEM I and $R^2 = 0.86$ for CEM III). These observations demonstrate that rheological parameters are linked to flowability indicators. BFA mobilizes more water and influences the mixture's viscosity, thus reducing the flowability of the pastes.

3.3. BFA water demand

The decrease in flowability and the modification of rheological properties of the pastes suggest a higher water demand in the pastes containing BFA. The standardized consistency of the pastes was determined using a Vicat apparatus in accordance with EN 196-3 [58]. Unlike the formulations in Tables 2 and 3 where a constant W/B ratio of 0.41 was used, the water content in this test was adjusted

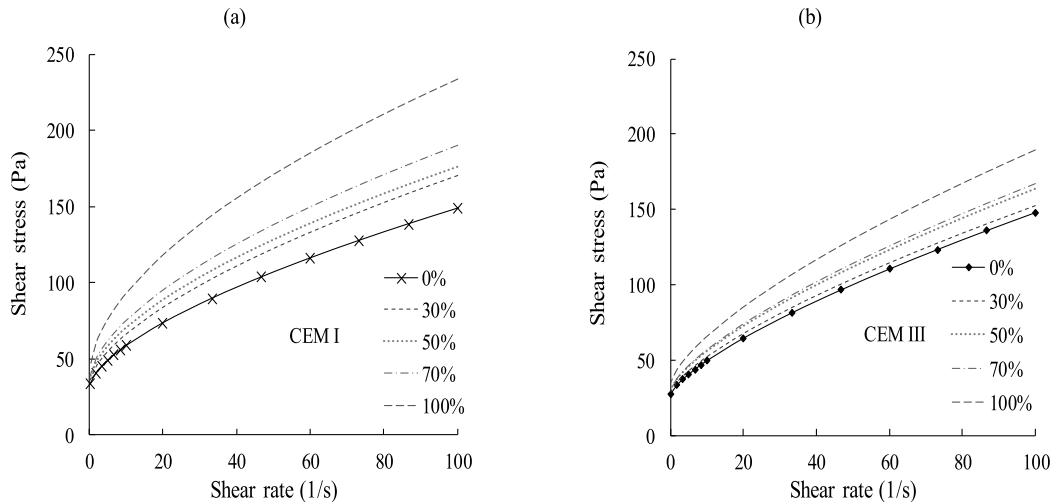


Fig. 6. Shear stress as a function of shear rate, pastes of (a) CEM I and (b) CEM III.

Table 4
Rheological properties of the pastes.

Pastes		τ_{HB} (Pa)	K_{HB} (Pa.s ⁿ)	n_{HB}
Cement	BFA content (%)			
CEM I	0	32.0	6.08	0.642
CEM I	30	34.1	7.65	0.625
CEM I	50	35.0	9.00	0.598
CEM I	70	36.9	9.81	0.597
CEM I	100	41.1	14.10	0.568
CEM III	0	27.0	4.31	0.724
CEM III	30	28.4	4.65	0.713
CEM III	50	29.6	5.24	0.704
CEM III	70	29.5	5.45	0.701
CEM III	100	34.0	6.50	0.689

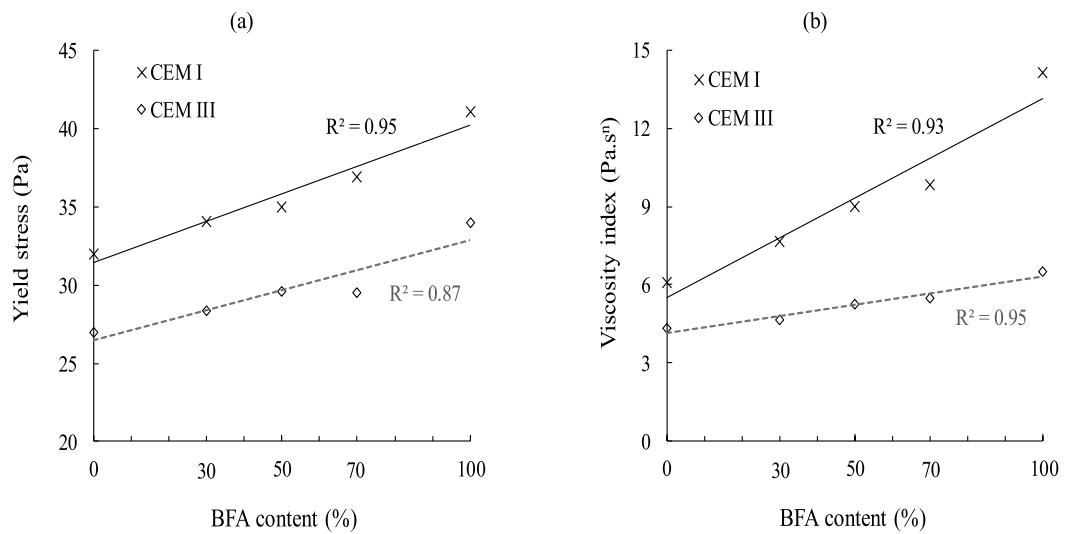


Fig. 7. Influence of BFA content on rheological characteristics of CEM I and CEM III pastes (a) Yield stress and (b) Viscosity index.

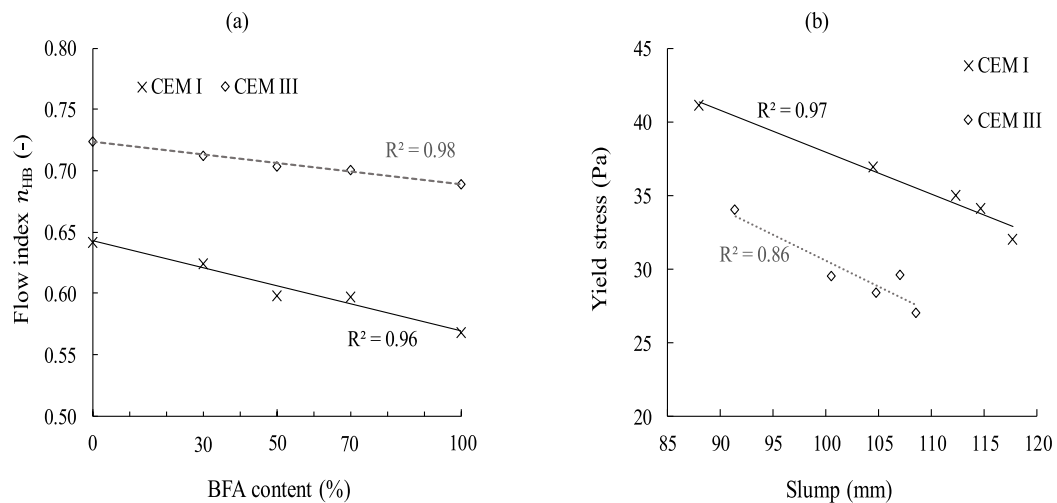


Fig. 8. (a) Influence of BFA content on flow index of CEM I and CEM III pastes and (b) correlation between slump and yield stress.

iteratively until the standardized consistency was reached for each substitution level, in order to quantify the additional water demand induced by BFA. For each substitution rate, the test was performed on three paste samples.

Thanks to the reference paste, composed of cement and filler, the quantity of water mobilized per unit mass of cement and filler is known for each mix. The ratio W/B represents the ratio of water to binder used to achieve the standardized consistency of the reference paste. In each mix, cement and filler will have a water demand corresponding to W/B times their present masses. The mass of limestone filler changes with each substitution, so the volume of water mobilized by the cement and limestone filler depends on the mass of limestone filler. The calculation of this quantity of water adsorbed by the cement/filler mixture is shown in (Eq. (2)).

$$\text{Water}_{\text{cement+filler}}(\text{g}) = (M_{\text{cement}} + M_{\text{filler}}) \times \frac{W}{B} \quad (2)$$

where M_{cement} and M_{filler} (g) are, the respective masses of cement and limestone filler present in the paste and W/B is, the ratio corresponding to the water demand of the mixture of cement and filler without BFA.

For each substitution rate, the water adsorbed by cement and filler in the mix is determined by applying (Eq. (2)). The amount of water adsorbed by the BFA is then derived from (Eq. (3)).

$$\text{Water}_{\text{BFA}}(\text{g}) = \text{Total water}(\text{g}) - \text{Water}_{\text{cement+filler}}(\text{g}) \quad (3)$$

where Total water (g) is the total amount of water in the paste for a standardized consistency.

The water demand generated by the BFA corresponds to the quantity of water required for the BFA. It is calculated using (Eq. (4)).

$$\text{Water demand}(\%) = \frac{\text{Water}_{\text{BFA}}}{M_{\text{BFA}}} \times 100 \quad (4)$$

with M_{BFA} being the mass of BFA in the paste (g).

The tests carried out with the Vicat apparatus show that the standardized consistency is achieved for increasing W/B ratios as a function of the substitution rate. The results of the measurements for the reference pastes and for all substitution rates are presented in Fig. 9. This increase in the amount of water necessary to reach the same consistency confirms that the water demand of BFA influences the properties of the pastes in the fresh state. Previous studies have shown an increase in the water demand of BFA due to the irregular shape of particles, the porosity, and the loss on ignition of the ashes [23]. The standardized consistency is reached for a W/B ratio of 0.301 and 0.335 for a CEM I + filler cement paste and a CEM III + filler cement paste, respectively.

Through (Eq. (2)), (Eq. (3)), and (Eq. (4)), the water demands generated by the substitution are determined and the results are detailed in Table 5. The progressive replacement of limestone filler with biomass fly ash increases the water demand required to achieve standardized consistency, mainly due to the coarser particle size, irregular morphology, and unburnt carbon of the ash, which enhance water absorption. This effect is less pronounced in pastes formulated with CEM III cement, likely because its finer particle size distribution and the vitreous surface of slag particles improve fluidity and reduce the sensitivity of the mixture to water content variations [54–56].

For the formulated self-compacting pastes, keeping a constant total water-to-binder ratio despite the observed water demand explains the reduction in flowability. A water adjustment can be made to the formulations of cementitious materials incorporating BFA to keep the effective water/binder ratio constant and ensure identical flowability in cases of substitution.

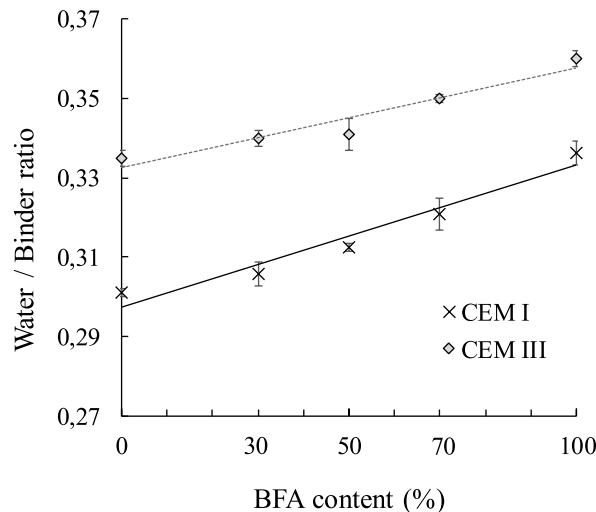


Fig. 9. Evolution of the W/B ratio to obtain the standardized consistency of pastes as a function of the substitution rate.

Table 5
Water demand generated by the BFA.

Pastes		Total water (g)	Water _{cement+filler} (g)	Water _{BFA} (g)	Water demand (%)
Cement	BFA content (%)				
CEM I	0	403	402	–	–
CEM I	30	411 ± 3.24	375.22	35.78 ± 3.24	40.20 ± 1.77
CEM I	50	418 ± 1.42	358.77	61.23 ± 1.42	41.09 ± 0.95
CEM I	70	432 ± 4.66	337.31	94.69 ± 4.66	45.09 ± 1.69
CEM I	100	451 ± 4.24	312.03	138.97 ± 4.24	46.71 ± 1.54
CEM III	0	445	444.00	–	–
CEM III	30	451 ± 2.12	412.72	38.78 ± 2.12	43.09 ± 2.36
CEM III	50	455 ± 1.41	392.29	62.72 ± 1.41	41.53 ± 0.94
CEM III	70	461 ± 1.41	371.85	89.15 ± 1.41	42.25 ± 0.67
CEM III	100	476 ± 0.71	341.03	135.47 ± 0.71	45.01 ± 0.23

3.4. Compressive strength

Compressive strengths were measured using a “RP8/250” FC (3R) hydraulic press. The mechanical properties of pastes were determined by compressive tests in accordance with the EN 196-1 standard for mortar testing, as no standardized protocol currently exists for mechanical testing of cement pastes [59]. After 7, 28 and 90 days, the pastes half test specimens (40 × 40 × 80 mm) are subjected to an increasing load of 2400 N/s until failure. The compressive strengths at 7, 28 and 90 days are presented in Fig. 10. As shown in Fig. 10a, pastes incorporating BFA exhibit similar strengths at 7 days (37 ± 1 MPa), slightly lower than those of the reference paste (40 ± 0.84 MPa). This reduction in strength corresponds to less than 9 % of the reference strength. At 28 days, a decrease in strength was observed with increasing substitution rates, with more pronounced differences in pastes incorporating biomass fly ash. This trend is likely related to the higher water demand associated with BFA, which can affect the packing density of the fines and the bond between particles [25]. However, by 90 days, the strength of pastes incorporating BFA reached 95 % of that of the reference paste, mirroring the trend observed at 7 days. While no studies specifically address the replacement of limestone filler with wood biomass fly ash, comparisons can be made with research where biomass ashes substitute cement or sand. In such cases, a more pronounced reduction in both early and long-term strength is often observed, mainly due to the lower reactivity of ash compared to clinker [24,46]. By contrast, the relatively modest strength loss observed here, along with later-age recovery, suggests that wood BFA primarily acts as an inert filler rather than a reactive component.

The compressive strengths measured on CEM III paste specimens are presented in Fig. 10b. A slight improvement in the 7-day strength is observed when wood BFA is used as a substitute for limestone filler in the CEM III pastes. At 28 days, all pastes incorporating BFA exhibit a strength of 51 ± 1 MPa, which is 2 MPa higher than the reference paste. At 90 days, the strengths of the pastes containing wood BFA show no notable differences compared to the reference pastes. The compressive strengths are considered to be relatively well maintained during substitution, as the increases are not significant.

The decrease in flowability is less pronounced in the formulations with CEM III compared to those with CEM I, which may also explain the subtle difference in behavior. Wood BFA interacts less physically with CEM III than with CEM I, which probably minimize its impact on compactness. The vitreous and reactive nature of slag provides favorable surface for both physical and potential chemical

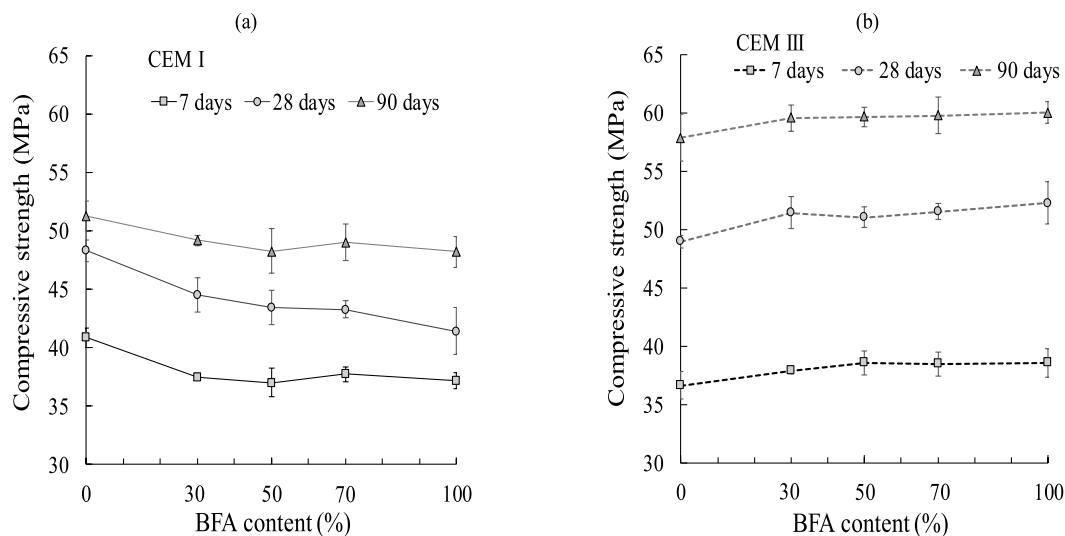


Fig. 10. Compressive strength at 7, 28 and 90 days of: (a) CEM I pastes and (b) CEM III pastes.

interactions with inert fillers such as wood biomass fly ash [56]. This results in a stronger BFA-slag interface in CEM III, improving microstructure and cohesion. The latent hydraulic reactivity of slag further contributes to the formation of additional hydrates, which enhance particle packing, and reinforce matrix cohesion [60,61]. This synergy explains why the incorporation of BFA has a smaller effect on the properties of CEM III pastes compared to CEM I.

3.5. Water-accessible porosity

The literature shows that the introduction of BFA into cementitious mixtures tends to increase porosity [40]. It is necessary to analyze the water-accessible porosity to assess the filling capacity of wood biomass fly ash fines and to understand the mechanical performances achieved. The water-accessible porosity is determined by hydrostatic weighing according to the French standard NF P18 459 on $4 \times 4 \times 8$ cm specimens [62]. The samples are placed into a vacuum chamber at a relative pressure of 25 mbar for 4 h to remove air from the pores before being immersed in water under vacuum for 44 h to achieve saturation. The water-accessible porosity (ϵ_{acc}) is then calculated using the equation (Eq. (5)):

$$\epsilon_{acc} = \frac{m_{air} - m_{dry}}{m_{air} - m_{water}} \quad (5)$$

with m_{air} (g) being the mass of the sample in saturated condition with dry surface, m_{water} (g) the mass of the saturated sample and m_{dry} (g) the mass of the dry specimen.

The water-accessible porosity of CEM I and CEM III pastes depending on BFA content are shown in Fig. 11. Previous studies have demonstrated that cement pastes are predominantly microporous materials and exhibit unique behavior in terms of water absorption. Indeed, the water-accessible porosity is always significantly higher than measurements reached with a helium pycnometer or mercury intrusion under dry conditions. The use of this measurement method in previous studies has shown that the determined values generally range between 27 % and 59 % [63–66]. Water, being a highly polar liquid, can infiltrate between the mineral layers of the material and increase the spacing between them. This creates new voids, leading to results that may appear elevated [63,64]. These results are, however, sufficiently representative to enable comparisons to be made.

Pastes formulated with CEM I and containing BFA have water-accessible porosities 3 % higher than those of reference pastes. The porosity remains relatively stable as the BFA substitution rate increases. At 90 days, porosity is lower and shows a slight tendency to decrease with increasing BFA content. The water demand of the ashes tends to influence the packing density of the fines, thereby increasing porosity. Additionally, the presence of certain chemical elements, such as magnesium oxides and lime, causes swelling during hydration, leading to increased porosity [46]. This higher porosity could partially explain the slight reduction in compressive strength observed in the CEM I pastes. Previous studies indicate that replacing cement with wood biomass fly ash generally leads to increased porosity due to the ash's low reactivity and relatively coarse particle size. Significant increases in water absorption and total porosity have been observed with wood BFA replacement levels ranging from 5 % to 70 %. Such changes are often associated with reduced mechanical strength and long-term durability [67,68]. However, when wood BFA is used as a filler substituting limestone, the increase in water-accessible porosity appears moderate (around 3 %) and tends to stabilize or decrease over time, especially at later curing ages. This behavior suggests that wood BFA acts primarily as an inert filler in this context, with limited impact on pore structure and mechanical performance.

At 90 days, CEM III pastes exhibit a slight decrease in water-accessible porosity with increasing BFA content. Their porosity is significantly lower than at 28 days, highlighting the role of slag hydration in long-term densification. In contrast, CEM I paste exhibit higher initial porosity and less reduction with curing, as Portland cement (OPC) lacks slag's long-term pore-refining capacity. The data

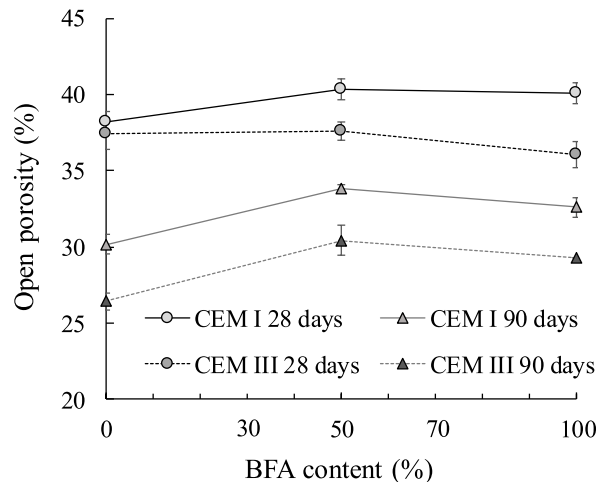


Fig. 11. Water-accessible porosity, at 28 and 90 days.

suggests that BFA's role is primarily physical, with no evidence of chemical contribution to porosity reduction.

3.6. Hydration kinetics

The morphology and fineness of BFA interact with the water in the paste and influence cement hydration. The impact of BFA on binder hydration was analyzed by measuring the heat of hydration of pastes over 48 h using an isothermal calorimeter "I-Cal 2000 HPX" in accordance with standard EN 196-11 [69]. Pastes samples were inserted into the calorimeter immediately after mixing and the temperature was kept constant at 20 °C during the tests to isolate the heat effect of hydration. The heat flows measured during the first 24 h of hydration and the heat quantities released over 48 h are shown in Fig. 12 for pastes formulated with CEM I.

Fig. 12a shows that replacing limestone filler with wood BFA leads to an increase in the heat of hydration between 6 and 12 h. During this period, pastes formulated with CEM I and BFA exhibit similar heat flow profiles, corresponding to the main hydration peak of C_3S in CEM I clinker and the formation of portlandite and CSH. The higher peak intensity observed with increasing BFA content suggests enhanced early formation of these hydrates. The cumulative heat released, as shown in Fig. 12b, also increases with BFA content, reaching a maximum at 70 % BFA, where the heat release is 34.02 J/g of cement higher than the reference paste. The difference in total heat released between 70 % and 100 % BFA substitution remains relatively small, indicating a threshold beyond which further BFA addition has limited effect.

Despite these variations in heat flow and cumulative heat, the overall hydration kinetics of CEM I pastes remain unchanged, with no significant shift in the timing of hydration peaks. The shape and position of the main hydration peak are preserved regardless of the BFA substitution rate, and no additional peaks are observed. This indicates that wood BFA, when used as a filler, does not induce the formation of additional hydrates or alter the fundamental reaction mechanisms of cement hydration.

This behavior is characteristic of an inert filler effect, where the introduction of fine particles enhances early hydration by providing more nucleation sites for hydrate formation, without participating in further chemical reactions. The increased early heat flow is thus attributed to physical effect rather than a chemical contribution from the BFA itself. This distinction confirms that wood BFA acts primarily as an inert filler in this context, facilitating the physical aspects of early hydration without modifying the overall hydration process. Specifically, BFA facilitates the creation of nucleation sites for hydrate phases [44,60,70].

The heat flow and cumulative heat released over 48 h for CEM III pastes are shown in Fig. 13. In Fig. 13a, the heat peak between 6 and 18 h represents the hydration reaction of C_3S and C_2S silicates in the clinker. The peak between 18 and 36 h reflects slag hydration. The slag in CEM III is rich in aluminum, silicon, and magnesium. This composition accelerates the hydration of calcium silicate phases and calcium aluminate phases. The substitution of limestone filler with BFA seems to intensify the hydration peak between 6 and 18 h. This may be due to increased nucleation sites from the higher specific surface area of BFA. From 18 to 36 h, the introduction of BFA delays and attenuates the peak progressively at 30 %, 50 % and 70 %. At 100 % BFA, the peak is eliminated. Thus, slag hydration is affected by the substitution of limestone filler. Previous studies have shown that fine limestone filler composed of calcite promotes carboaluminates formation at early ages, which accelerates and amplifies the reaction [71]. Limestone filler gives the slag a slightly alkaline environment, enhancing the activation of slag particles [61,72–74]. The increase in viscosity observed previously also influences the diffusion of ions necessary for slag activation. The heat-release curves shown in Fig. 13b confirm these observations. The more intense early hydration reactions in pastes with BFA result in a slight increase in the amount of heat released.

In all pastes incorporating BFA, both with CEM I and CEM III, the chemical composition differences between wood BFA and

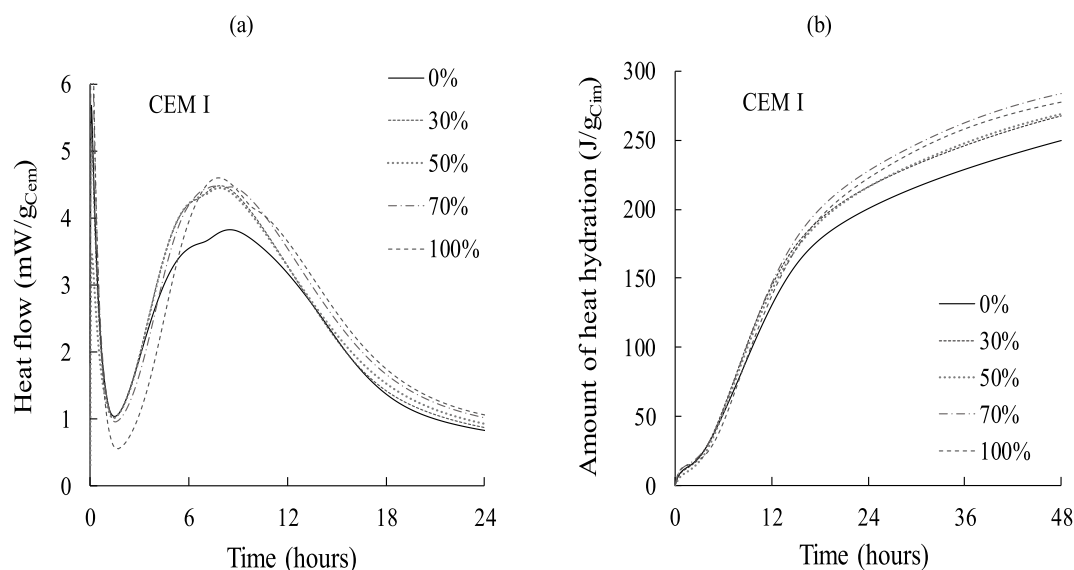


Fig. 12. CEM I pastes: (a) Heat flow and (b) amount of heat hydration, as a function of time.

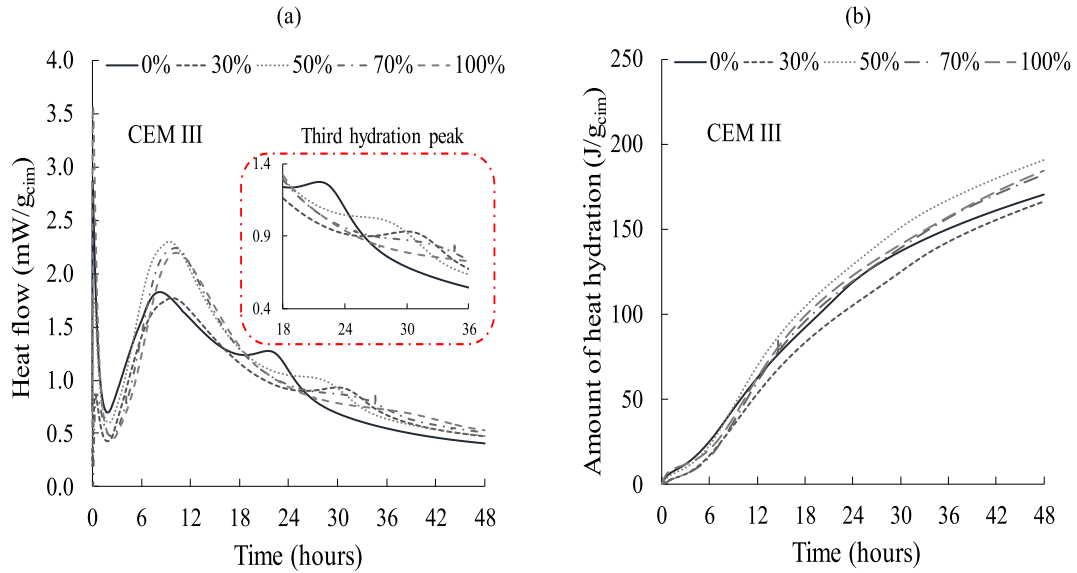


Fig. 13. CEM III pastes: (a) heat flow and (b) amount of heat hydration, as a function of time.

limestone filler influence the hydration processes. However, in the CEM III-based pastes, the water demand of BFA and the reduction in the amount of limestone filler appear to slow down the early-age activation and hydration of slag.

3.7. Thermogravimetric analysis

The hydration kinetics indicate increased hydrate production during the initial hours. Thermogravimetric analyses were conducted to evaluate the influence of BFA on long-term hydrate production. These analyses were done with a NETZSCH STA 2500 Regulus analyzer at 7 and 90 days. Samples were prepared from strength test specimens. Fragments from the specimen cores, unexposed to carbonation, were manually ground and sieved to a particle size of 400 μm . Approximately 140 ± 10 mg of each sample was analyzed in a nitrogen atmosphere (50 ml/min) to prevent undesirable reactions during measurement. The temperature range was set from 25 $^{\circ}\text{C}$ to 1000 $^{\circ}\text{C}$, with a heating rate of 10 $^{\circ}\text{C}$ per minute.

Figs. 14 and 15 show the derivative thermogravimetric (DTG) curves at 7 days and 90 days of curing as a function of temperature. The mass loss around 130 $^{\circ}\text{C}$ represents the dehydration of ettringite and the evaporation of part of the water from the carboaluminates. The peak representing CSH dehydration is between 150 $^{\circ}\text{C}$ and 180 $^{\circ}\text{C}$. The decomposition of portlandite $\text{Ca}(\text{OH})_2$ is identified by the peak between 400 $^{\circ}\text{C}$ and 500 $^{\circ}\text{C}$. Between 600 $^{\circ}\text{C}$ and 800 $^{\circ}\text{C}$, the peak is associated with the decomposition of calcite CaCO_3 [70].

Observation of the DTG curves in Figs. 14 and 15 shows that the BFA incorporated mainly acted as an inert filler within the cement matrix. The absence of any noticeable change in the intensity of portlandite peak across all substitution rates suggests that BFA does not interfere with the key hydration processes. In the longer term, this trend persists with the DTG profiles remaining unchanged regardless of BFA content. For CEM I pastes at 7 days, a slight variation is observed in the amount of ettringite and CSH. Similarly, for CEM III pastes, at 7 and 90 days, BFA does not affect major hydration phases. The characteristic peak corresponding to portlandite remain consistent.

Compared to other biomass fly ashes, such as rice husk ash (RHA) or sugarcane bagasse ash (SCBA), which may exhibit partial pozzolanic reactivity due to their higher amorphous silica content, wood-based BFA behaves differently. Studies have shown that RHA and SCBA can gradually consume portlandite and contribute to the formation of secondary CSH, thereby influencing the hydration kinetics of the cement matrix [38,75]. In contrast, the wood BFA used in this study shows no evidence of such reactivity, indicating that its chemical composition and mineralogy lead to a primarily inert behavior. As such, the observed behavior of wood BFA aligns more closely with that of inert mineral fillers, rather than reactive bio-ashes.

The mass contents of portlandite and calcite were determined to quantify the influence of wood BFA on cement hydration. Comparing these phases between mixes allows for a clearer distinction between physical filler effects and potential chemical contributions from BFA. The thermogravimetric analysis (TGA) was used to quantify portlandite and calcite based on equations (Eq. (6)) and (Eq. (7)) respectively:

$$\%_{\text{Ca}(\text{OH})_2} = \frac{\Delta m_{400^{\circ}\text{C}-550^{\circ}\text{C}}}{m_c} \times \frac{M_{\text{Ca}(\text{OH})_2}}{M_{\text{H}_2\text{O}}} \quad (6)$$

$$\%_{\text{CaCO}_3} = \frac{\Delta m_{600^{\circ}\text{C}-1000^{\circ}\text{C}}}{m_c} \times \frac{M_{\text{CaCO}_3}}{M_{\text{CO}_2}} \quad (7)$$

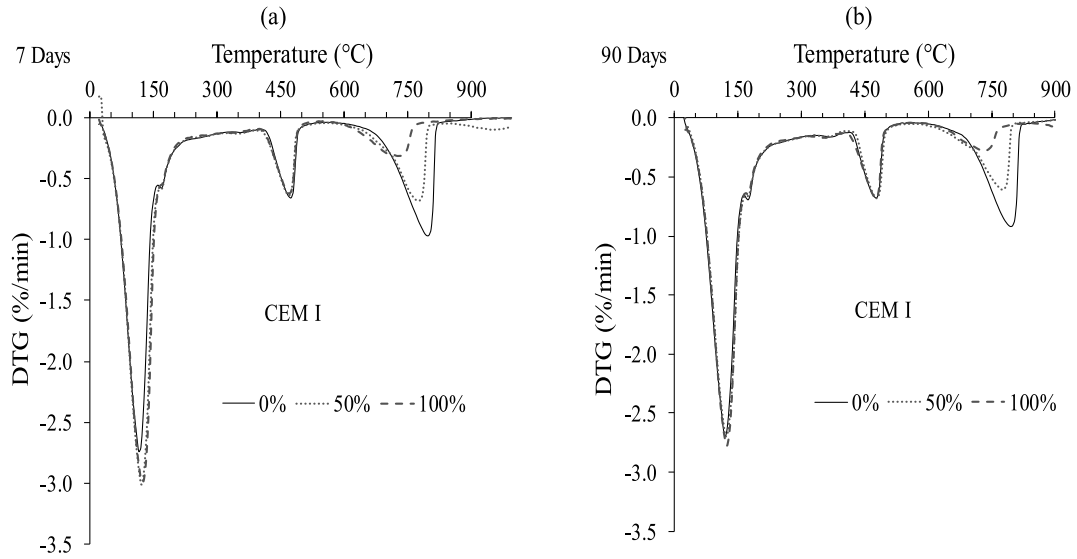


Fig. 14. DTG as a function of temperature. CEM I pastes: (a) 7 days and (b) 90 days.

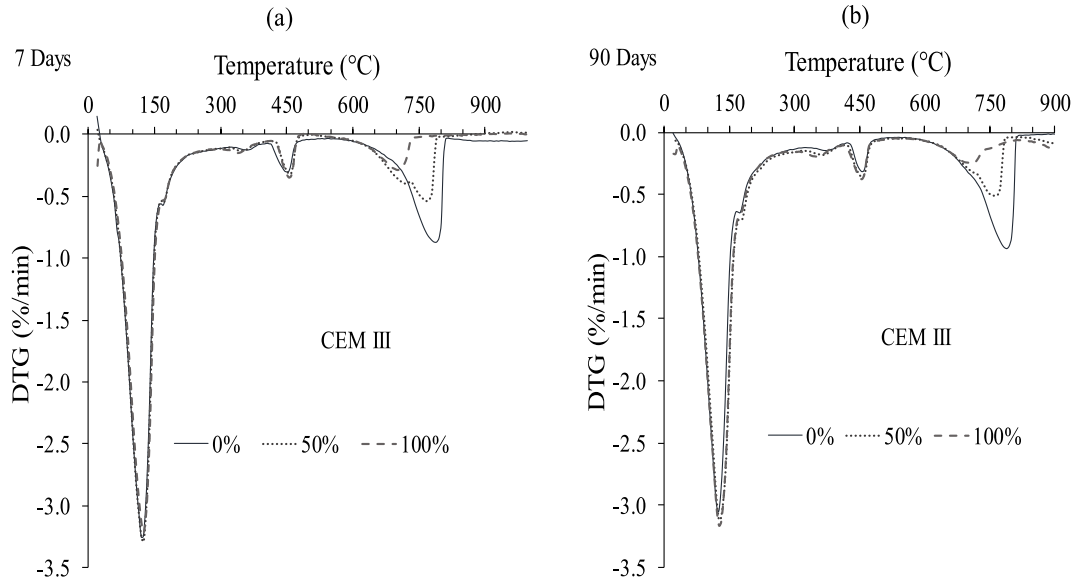


Fig. 15. DTG as a function of temperature. CEM III pastes: (a) 7 days and (b) 90 days.

With $\Delta m_{400^{\circ}\text{C}-550^{\circ}\text{C}}$ the mass loss of the sample between 400 and 550 °C, $\Delta m_{600^{\circ}\text{C}-1000^{\circ}\text{C}}$ the mass loss of the sample between 600 °C and 1000 °C and $M_{\text{Ca}(\text{OH})_2}$ (74 g/mol), M_{CaCO_3} (100 g/mol), $M_{\text{H}_2\text{O}}$ (18 g/mol) and M_{CO_2} (44 g/mol) the molar masses of portlandite, calcite, water and CO_2 respectively. m_c the mass of anhydrous cement, is obtained by (Eq. (8))

$$m_c = \frac{m_{\text{sample}}}{(1 + W/C) \times (1 + \text{LOI})} \quad (8)$$

Where W/C is the water/cement mass ratio and LOI the loss on ignition of the cement (2.7 % for CEM I and 2.3 % for CEM III/A).

Another critical indicator of microstructural evolution in cementitious pastes is the degree of hydration, which quantifies the fraction of clinker that has reacted with water to form hydrates. In this study, the degree of hydration was determined using thermogravimetric analysis (TGA), calculated from mass losses in accordance with (Eq. (9)) [76]:

$$\alpha = \frac{m_{\text{wb}}}{W_{\text{el}} \times m_c} \times 100 \quad (9)$$

With W_{el} the quantity of water required for complete hydration of the cement, 23.5 % for CEM I and 24 % for CEM III/A and m_{wb} the mass of cement-bound water in the sample, calculated by equation (Eq. (10)).

$$m_{wb} = \Delta m_{145-1000} - \Delta m_{550-800} - m_c \times LOI \quad (10)$$

The obtained values are presented in Table 6. The results indicate relatively minor variations in portlandite content and degree of hydration as a function of biomass fly ash (BFA) content. In CEM I pastes, at 90 days, the portlandite content ranges from 24.71 % (0 % BFA) to 24.84 % (50 % BFA) and 23.01 % (100 % BFA), showing a variation of less than 2 %. The degree of hydration remained generally stable across the different CEM I formulations, with values ranging from 50.53 % to 53.89 % at 7 days and 65.64 %–67.76 % at 90 days. These results suggest that the introduction of BFA does not have a significant effect on hydrate development in these mixes. In CEM III pastes, the observed variations are slightly more pronounced, particularly for the degree of hydration at 90 days, which increases from 48.75 % (0 % BFA) to 61.06 % (50 % BFA) and 58.41 % (100 % BFA). These increases can be attributed to physical changes to the microstructure, such as a modification in capillary porosity, consistent with the porosity measurements. The gradual decrease in calcite peak intensity is due to the reduction of limestone filler in the formulations. Limestone filler is composed of 98 % carbonates. This explains the greater mass loss when the limestone filler content is higher.

Wood BFA primarily interacts physically with the matrix through physical mechanisms, similarly to limestone filler. These findings align with the mechanical results, particularly the compressive strength, which is generally maintained in BFA-containing pastes. Due to its physical properties, BFA appears to function comparably to limestone filler in the long-term formation of CSH and portlandite.

4. Conclusion

This study evaluated the possibility of using biomass fly ash as a substitute for limestone filler, through a study on paste. The analysis of the physicochemical properties of biomass fly ash revealed that it consists of fine, non-reactive particles and can be considered as an inert mineral addition. The main observations made are.

- (1) The presence of unburnt residues in wood BFA results in a higher water demand compared to limestone filler. The hypothesis that BFA requires more water is confirmed by determining the W/B ratio necessary to achieve a standardized consistency for each substitution level. The test showed a change in the ratio with the BFA content, proving that it generates a higher water demand.
- (2) The use of BFA as a substitute for in filler substitution results in a decrease in flowability that increases with the substitution rate. Flowability can be conserved by water correction or the use of superplasticizer.
- (3) The incorporation of biomass fly ash into the paste increases the viscosity and yield stress of the pastes in line with the substitution rate. This is due to increased friction between the solid particles. Pastes show increasingly shear-thinning behavior with the substitution rate.
- (4) Compressive strengths are relatively similar during in cases of substitution. BFA pastes have the same strength regardless of the substitution rate. Compressive strengths should not be an obstacle to substitution.
- (5) Slag particles in CEM III probably improve material cohesion through better adhesion with BFA. They reduce the impact of BFA on paste compactness and on interactions between fines. The smaller lesser effect of BFA on the properties of CEM III pastes compared to CEM I pastes may be attributed to this phenomenon.

In future work, a water adjustment or superplasticizer would be applied to the formulations to keep the effective water content and flowability constant. The observed interactions between the BFA and the cementitious matrix are physical and highlight the filler effect of wood biomass fly ash. Their use as a replacement for limestone filler is feasible. Since wood biomass fly ash has a linear influence on paste properties, a 100 % substitution of limestone filler with BFA is planned for the continuation of the study on the scale of concretes. While this study focused on rheological and mechanical properties, further investigation is required into the durability performance of BFA cementitious materials, notably with respect to carbonation resistance, chloride ingress, and freeze-thaw resistance, which are key for their practical application.

Table 6
Portlandite and calcite content by mass and degree of hydration of pastes.

Pastes		%Ca(OH) ₂ (%)		%CaCO ₃ (%)		α (%)	
Cement	BFA content (%)	7 days	90 days	7 days	90 days	7 days	90 days
CEM I	0	24.71	27.18	31.07	30.79	51.84	65.64
CEM I	50	24.84	27.71	26.89	28.53	53.89	65.36
CEM I	100	23.01	27.92	12.67	17.35	50.53	67.76
CEM III	0	12.03	12.04	31.06	30.92	39.38	48.75
CEM III	50	13.36	13.89	25.60	21.35	39.72	61.06
CEM III	100	9.19	14.34	8.28	15.60	36.91	58.41

CRedit authorship contribution statement

Joëlla Grâce Dossa: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Jonathan Page:** Writing – review & editing, Methodology, Conceptualization. **Yannick Vanhove:** Writing – review & editing, Methodology, Conceptualization. **Chafika Djelal:** Writing – review & editing, Validation, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] J. Zhai, I.T. Burke, D.I. Stewart, Beneficial management of biomass combustion ashes, *Renew. Sustain. Energy Rev.* 151 (2021) 111555, <https://doi.org/10.1016/j.rser.2021.111555>.
- [2] I. Carević, M. Serdar, N. Štirmer, N. Ukrainczyk, Preliminary screening of wood biomass ashes for partial resources replacements in cementitious materials, *J. Clean. Prod.* 229 (2019) 1045–1064, <https://doi.org/10.1016/j.jclepro.2019.04.321>.
- [3] C. Venkata Sudhakar, G. Umamaheswara Reddy, N. Usha Rani, Delineation and evaluation of the captive limestone mining area change and its influence on the environment using multispectral satellite images for industrial long-term sustainability, *Clean. Eng. Technol.* 10 (2022) 100551, <https://doi.org/10.1016/j.clet.2022.100551>.
- [4] N. Li, Y. Guo, L. Wang, Q. Wang, D. Yan, S. Zhao, T. Lei, Evaluation and quantitative characterization for the ecological environment impact of open pit mining on vegetation destruction from landsat time series: a case study of Wulishan limestone mine, *Ecol. Indic.* 158 (2024) 111371, <https://doi.org/10.1016/j.ecolind.2023.111371>.
- [5] MdM.H. Khan, MdH.R. Sobuz, M.M. Meraz, V.W.Y. Tam, N.MdS. Hasan, N.M.N. Shaurdho, Effect of various powder content on the properties of sustainable self-compacting concrete, *Case Stud. Constr. Mater.* 19 (2023) e02274, <https://doi.org/10.1016/j.cscm.2023.e02274>.
- [6] Y. Shi, N. Feng, T. Hao, Influence of ultrafine powder on the fluidity and strength of cement paste, *Adv. Cem. Res. - ADV CEM RES* 12 (2000) 89–95, <https://doi.org/10.1680/adcr.2000.12.3.89>.
- [7] M. Nehdi, Microfiller Effect on Rheology, Microstructure, and Mechanical Properties of high-performance Concrete, University of British Columbia, 1998, <https://doi.org/10.14288/1.0050312>.
- [8] H. Moosberg-Bustnes, The function of fillers in concrete, *Mater. Struct.* 37 (2004) 74–81, <https://doi.org/10.1617/13694>.
- [9] EFFC/DFI Guide to Tremie Concrete for Deep Foundations 2nd Edition, EFFC (n.d.), https://www.effc.org/media_corner/effc-dfi-guide-to-tremie-concrete-for-deep-foundations-2nd-edition/ (accessed September 9, 2024).
- [10] European guidelines for self compacting concrete (SCC) | EFCA, n.d. <https://www.efca.info/download/european-guidelines-for-self-compacting-concrete-scc/>. (Accessed 9 September 2024).
- [11] F. Han, S. Pu, Y. Zhou, H. Zhang, Z. Zhang, Effect of ultrafine mineral admixtures on the rheological properties of fresh cement paste: a review, *J. Build. Eng.* 51 (2022) 104313, <https://doi.org/10.1016/j.jobbe.2022.104313>.
- [12] S.V. Vassilev, D. Baxter, L.K. Andersen, C.G. Vassileva, An overview of the chemical composition of biomass, *Fuel* 89 (2010) 913–933, <https://doi.org/10.1016/j.fuel.2009.10.022>.
- [13] R. Siddique, Utilization of wood ash in concrete manufacturing, *Resour. Conserv. Recycl.* 67 (2012) 27–33, <https://doi.org/10.1016/j.resconrec.2012.07.004>.
- [14] C.B. Cheah, M. Ramli, The implementation of wood waste ash as a partial cement replacement material in the production of structural grade concrete and mortar: an overview, *Resour. Conserv. Recycl.* 55 (2011) 669–685, <https://doi.org/10.1016/j.resconrec.2011.02.002>.
- [15] M. Berra, T. Mangialardi, A.E. Paolini, Reuse of woody biomass fly ash in cement-based materials, *Constr. Build. Mater.* 76 (2015) 286–296, <https://doi.org/10.1016/j.conbuildmat.2014.11.052>.
- [16] L.A.C. Tarelho, E.R. Teixeira, D.F.R. Silva, R.C.E. Modolo, J.A. Labrincha, F. Rocha, Characteristics of distinct ash flows in a biomass thermal power plant with bubbling fluidised bed combustor, *Energy* 90 (2015) 387–402, <https://doi.org/10.1016/j.energy.2015.07.036>.
- [17] A. Danish, O. Karadag, T. Bilir, T. Ozbakkaloglu, Valorization of biomass ashes in the production of cementitious composites: a comprehensive review of properties and performance, *Constr. Build. Mater.* 405 (2023) 133244, <https://doi.org/10.1016/j.conbuildmat.2023.133244>.
- [18] J.P. Moretti, S. Nunes, A. Sales, Self-compacting concrete incorporating sugarcane bagasse ash, *Constr. Build. Mater.* 172 (2018) 635–649, <https://doi.org/10.1016/j.conbuildmat.2018.03.277>.
- [19] M. Jamil, A.B.M.A. Kaish, S.N. Raman, M.F.M. Zain, Pozzolan contribution of rice husk ash in cementitious system, *Constr. Build. Mater.* 47 (2013) 588–593, <https://doi.org/10.1016/j.conbuildmat.2013.05.088>.
- [20] H.Bt Awang, M.Z. Al-Mulali, The inclusion of palm oil ash biomass waste in concrete: a literature review, in: V. Waisundara (Ed.), *Palm Oil, InTech*, 2018, <https://doi.org/10.5772/intechopen.76632>.
- [21] M.A. Megat Johari, A.M. Zeyad, N. Muhamad Bunnori, K.S. Ariffin, Engineering and transport properties of high-strength green concrete containing high volume of ultrafine palm oil fuel ash, *Constr. Build. Mater.* 30 (2012) 281–288, <https://doi.org/10.1016/j.conbuildmat.2011.12.007>.

- [22] S. Demis, J.G. Tapali, V.G. Papadakis, An investigation of the effectiveness of the utilization of biomass ashes as pozzolanic materials, *Constr. Build. Mater.* 68 (2014) 291–300, <https://doi.org/10.1016/j.conbuildmat.2014.06.071>.
- [23] S. Wang, A. Miller, E. Llamazos, F. Fonseca, L. Baxter, Biomass fly ash in concrete: mixture proportioning and mechanical properties, *Fuel* 87 (2008) 365–371, <https://doi.org/10.1016/j.fuel.2007.05.026>.
- [24] E.R. Teixeira, A. Camões, F.G. Branco, Valorisation of wood fly ash on concrete, *Resour. Conserv. Recycl.* 145 (2019) 292–310, <https://doi.org/10.1016/j.resconrec.2019.02.008>.
- [25] R. Rajamma, L. Senff, M.J. Ribeiro, J.A. Labrincha, R.J. Ball, G.C. Allen, V.M. Ferreira, Biomass fly ash effect on fresh and hardened state properties of cement based materials, *Composites, Part B* 77 (2015) 1–9, <https://doi.org/10.1016/j.compositesb.2015.03.019>.
- [26] S. Kekez, I. Nováková, M. Lach, K. Setlak, L. Fiala, J. Příkryl, G. Furtos, I. Petean, K.F. Alexandersson, J. Baronins, I. Hager, Waste valorization for fabrication of geopolymers, *Case Stud. Constr. Mater.* 22 (2025) e04388, <https://doi.org/10.1016/j.cscm.2025.e04388>.
- [27] R. Kaminskas, V. Cesnauskas, Influence of activated biomass fly ash on portland cement hydration, *Ceram. Silik.* 58 (2014) 260–268.
- [28] I. Gabrijel, M. Jelčić Rukavina, N. Štirmer, Influence of wood fly ash on concrete properties through filling effect mechanism, *Materials* 14 (2021) 7164, <https://doi.org/10.3390/ma14237164>.
- [29] J. Cuenca, J. Rodríguez, M. Martín-Morales, Z. Sánchez-Roldán, M. Zamorano, Effects of olive residue biomass fly ash as filler in self-compacting concrete, *Constr. Build. Mater.* 40 (2013) 702–709, <https://doi.org/10.1016/j.conbuildmat.2012.09.101>.
- [30] P. Huang, B. Huang, J. Li, N. Wu, Q. Xu, Application of sugar cane bagasse ash as filler in ultra-high performance concrete, *J. Build. Eng.* 71 (2023) 106447, <https://doi.org/10.1016/j.jobe.2023.106447>.
- [31] V.G. Jiménez-Quero, F.M. León-Martínez, P. Montes-García, C. Gaona-Tiburcio, J.G. Chacón-Nava, Influence of sugar-cane bagasse ash and fly ash on the rheological behavior of cement pastes and mortars, *Constr. Build. Mater.* 40 (2013) 691–701, <https://doi.org/10.1016/j.conbuildmat.2012.11.023>.
- [32] G. Athira, A. Bahurudeen, Rheological properties of cement paste blended with sugarcane bagasse ash and rice straw ash, *Constr. Build. Mater.* 332 (2022) 127377, <https://doi.org/10.1016/j.conbuildmat.2022.127377>.
- [33] J. Šantek Bajto, N. Štirmer, S. Cerковиć, I. Carević, K. Kostanić Jurić, Pilot Scale production of precast concrete elements with wood biomass ash, *Materials* 14 (2021) 6578, <https://doi.org/10.3390/ma14216578>.
- [34] M.A. Azzi, Etude du ressage des bétons de parois moulées - influence des paramètres de formulation, Thèse de doctorat, Artois, 2017. <https://theses.fr/2017ARTO0204>. (Accessed 9 September 2024).
- [35] A.R.E. Zein, Etude des mécanismes des cheminées de remontée d'eau des parois moulées, thesis, Artois, <https://theses.fr/s274030>, 2023. (Accessed 17 June 2025).
- [36] F. Schlupp, J. Page, C. Djelal, L. Libessart, Use of biomass bottom ash as granular substitute in mortar, *J. Build. Eng.* 75 (2023) 106927, <https://doi.org/10.1016/j.jobe.2023.106927>.
- [37] S. Wang, L. Baxter, Comprehensive study of biomass fly ash in concrete: strength, microscopy, kinetics and durability, *Fuel Process. Technol.* 88 (2007) 1165–1170, <https://doi.org/10.1016/j.fuproc.2007.06.016>.
- [38] G.C. Cordeiro, R.D. Toledo Filho, L.M. Tavares, E.M.R. Fairbairn, Pozzolanic activity and filler effect of sugar cane bagasse ash in Portland cement and lime mortars, *Cem. Concr. Compos.* 30 (2008) 410–418, <https://doi.org/10.1016/j.cemconcomp.2008.01.001>.
- [39] I. Carević, N. Štirmer, M. Serdar, N. Ukrainczyk, Effect of wood biomass ash storage on the properties of cement composites, *Materials* 14 (2021) 1632, <https://doi.org/10.3390/ma14071632>.
- [40] X. Liang, Z. Li, H. Dong, G. Ye, A review on the characteristics of wood biomass fly ash and their influences on the valorization in cementitious materials, *J. Build. Eng.* 97 (2024) 110927, <https://doi.org/10.1016/j.jobe.2024.110927>.
- [41] S.V. Vassilev, D. Baxter, L.K. Andersen, C.G. Vassileva, An overview of the composition and application of biomass ash. Part 1. Phase-mineral and chemical composition and classification, *Fuel* 105 (2013) 40–76, <https://doi.org/10.1016/j.fuel.2012.09.041>.
- [42] R. Rumman, M.R. Kamal, A. Bediwy, M.S. Alam, Partially burnt wood fly ash characterization and its application in low-carbon mortar and concrete, *Constr. Build. Mater.* 402 (2023) 132946, <https://doi.org/10.1016/j.conbuildmat.2023.132946>.
- [43] J.-M. Lessard, A. Omran, A. Tagnit-Hamou, R. Gagne, Feasibility of using biomass fly and bottom ashes in dry-cast concrete production, *Constr. Build. Mater.* 132 (2017) 565–577, <https://doi.org/10.1016/j.conbuildmat.2016.12.009>.
- [44] B. Lothenbach, K. Scrivener, R.D. Hooton, Supplementary cementitious materials, *Cement Concr. Res.* 41 (2011) 1244–1256, <https://doi.org/10.1016/j.cemconres.2010.12.001>.
- [45] H. Sun, L. Pang, Y. Ding, B. Xing, Y. Tang, X. Sun, J. Yuan, L. Zhou, Y. Wang, Y. Lyu, F. Sha, Influence of the physical morphological characteristics of mineral fillers on the bitumen-filler interfacial interaction, *Constr. Build. Mater.* 378 (2023) 131206, <https://doi.org/10.1016/j.conbuildmat.2023.131206>.
- [46] N. Ukrainczyk, Reuse of woody biomass ash waste in cementitious materials, *Chem. Biochem. Eng. Q.* 30 (2016) 137–148, <https://doi.org/10.15255/CABEQ.2015.2231>.
- [47] G.M. Nunes, M.A.S. Anjos, A.B.S.M. Lins, A.M.S. Negreiros, L.R. Pessoa, Evaluation of the mechanical behaviour of representative volumetric elements of 3DCP masonry mixtures with partial replacement of cement by limestone filler and metakaolin, *J. Build. Eng.* 78 (2023) 107650, <https://doi.org/10.1016/j.jobe.2023.107650>.
- [48] G. Rojo-López, B. González-Fontebona, F. Martínez-Abella, I. González-Taboada, Rheology, durability, and mechanical performance of sustainable self-compacting concrete with metakaolin and limestone filler, *Case Stud. Constr. Mater.* 17 (2022) e01143, <https://doi.org/10.1016/j.cscm.2022.e01143>.
- [49] R. Rajamma, R.J. Ball, L.A.C. Tarelho, G.C. Allen, J.A. Labrincha, V.M. Ferreira, Characterisation and use of biomass fly ash in cement-based materials, *J. Hazard. Mater.* 172 (2009) 1049–1060, <https://doi.org/10.1016/j.jhazmat.2009.07.109>.
- [50] M. Haist, J. Link, D. Nicia, S. Leinitz, C. Baumert, T. Bronk, D. Cotardo, M. Eslami Pirharati, S. Fataei, H. Garrecht, C. Gehlen, I. Hauschildt, I. Ivanova, S. Jesinghausen, C. Klein, H.-W. Krauss, L. Lohaus, D. Lowke, O. Mazanec, V. Mechtcherine, Interlaboratory study on rheological properties of cement pastes and reference substances: comparability of measurements performed with different rheometers and measurement geometries, *Mater. Struct.* 53 (2020), <https://doi.org/10.1617/s11527-020-01477-w>.
- [51] E.R. Teixeira, R. Mateus, A. Camões, F.G. Branco, Quality and durability properties and life-cycle assessment of high volume biomass fly ash mortar, *Constr. Build. Mater.* 197 (2019) 195–207, <https://doi.org/10.1016/j.conbuildmat.2018.11.173>.
- [52] Effects of the incorporation of sawdust waste incineration fly ash in cement pastes and mortars, (n.d.). <https://doi.org/10.3130/jaabe.3.1>.
- [53] C. Chandara, E. Sakai, K.A.M. Azizli, Z.A. Ahmad, S.F.S. Hashim, The effect of unburned carbon in palm oil fuel ash on fluidity of cement pastes containing superplasticizer, *Constr. Build. Mater.* 24 (2010) 1590–1593, <https://doi.org/10.1016/j.conbuildmat.2010.02.036>.
- [54] J.-H. Lim, G. Lee, Effect of blast furnace slag on rheological properties of fresh mortar, *J. Archit. Inst. Korea Struct. Constr.* 30 (2014) 51–58, <https://doi.org/10.5659/JAIK.SC.2014.30.6.51>.
- [55] Y. Oinam, S. Ju, S. Gwon, M. Shin, S. Pyo, Characteristics of GGBFS-based pervious concrete considering rheological properties of the binder, *Int. J. Concr. Struct. Mater.* 16 (2022), <https://doi.org/10.1186/s40069-022-00551-7>.
- [56] Y.-X. Shi, I. Matsui, Y.-J. Guo, A study on the effect of fine mineral powders with distinct vitreous contents on the fluidity and rheological properties of concrete, *Cement Concr. Res.* 34 (2004) 1381–1387, <https://doi.org/10.1016/j.cemconres.2003.12.031>.
- [57] V.-H. Nguyen, S. Remond, J.-L. Gallias, Influence of cement grouts composition on the rheological behaviour, *Cement Concr. Res.* 41 (2011) 292–300, <https://doi.org/10.1016/j.cemconres.2010.11.015>.
- [58] NF EN 196-3, afnor Ed, n.d. <https://www.boutique.afnor.org/fr-fr/norme/nf-en-1963/methodes-dessai-des-ciments-partie-3-determination-du-temps-de-prise-et-de-fa186374/79422>. (Accessed 10 September 2025).
- [59] NF EN 196-1, afnor Ed, n.d. <https://www.boutique.afnor.org/fr-fr/norme/nf-en-1961/methodes-dessai-des-ciments-partie-1-determination-des-resistances/fa184622/57803>. (Accessed 12 September 2025).
- [60] W.A. Gutteridge, J.A. Dalziel, Filler cement: the effect of the secondary component on the hydration of Portland cement: part I. A fine non-hydraulic filler, *Cement Concr. Res.* 20 (1990) 778–782, [https://doi.org/10.1016/0008-8846\(90\)90011-L](https://doi.org/10.1016/0008-8846(90)90011-L).

- [61] P. Mounanga, M.I.A. Khokhar, R. El Hachem, A. Loukili, Improvement of the early-age reactivity of fly ash and blast furnace slag cementitious systems using limestone filler, *Mater. Struct.* 44 (2011) 437–453, <https://doi.org/10.1617/s11527-010-9637-1>.
- [62] NF P18-459, afnor Ed, n.d. <https://www.boutique.afnor.org/fr-fr/norme/nf-p18459/beton-essai-pour-beton-durci-essai-de-porosite-et-de-masse-volumique/fa160729/34961>. (Accessed 12 September 2025).
- [63] M. Krus, K.K. Hansen, H.M. Künzel, Porosity and liquid absorption of cement paste, *Mater. Struct.* 30 (1997) 394–398, <https://doi.org/10.1007/BF02498561>.
- [64] R.L. Day, B.K. Marsh, Measurement of porosity in blended cement pastes, *Cement Concr. Res.* 18 (1988) 63–73, [https://doi.org/10.1016/0008-8846\(88\)90122-6](https://doi.org/10.1016/0008-8846(88)90122-6).
- [65] A. Bordy, A. Younsi, B. Fiorio, S. Aggoun, Hydratation de matériaux cimentaires à base, d'une fine issue du recyclage d'une pâte de ciment (2015), <https://doi.org/10.13140/RG.2.1.4442.8249>.
- [66] A. Lobet, Influence des paramètres de composition des matériaux cimentaires sur les propriétés de transfert, These de doctorat, Toulouse, INSA, 2003. <https://theses.fr/2003ISAT0016>. (Accessed 8 October 2024).
- [67] J. Fořt, J. Šál, R. Ševčík, M. Doleželová, M. Keppert, M. Jerman, M. Záleská, V. Stehel, R. Černý, Biomass fly ash as an alternative to coal fly ash in blended cements: functional aspects, *Constr. Build. Mater.* 271 (2021) 121544, <https://doi.org/10.1016/j.conbuildmat.2020.121544>.
- [68] S. Maschio, G. Tonello, L. Piani, E. Furlani, Fly and bottom ashes from biomass combustion as cement replacing components in mortars production: rheological behaviour of the pastes and materials compression strength, *Chemosphere* 85 (2011) 666–671, <https://doi.org/10.1016/j.chemosphere.2011.06.070>.
- [69] NF EN 196-11, Afnor Ed. (n.d.). <https://www.boutique.afnor.org/fr-fr/norme/nf-en-19611/methodes-dessais-des-ciments-partie-11-chaleur-dhydratation-methode-par-cal/fa189490/83184> (accessed September 12, 2025).
- [70] K. Scrivener, R. Snellings, B. Lothenbach, *A Practical Guide to Microstructural Analysis of Cementitious Materials*, 2015.
- [71] D. Wang, C. Shi, N. Farzadnia, Z. Shi, H. Jia, Z. Ou, A review on use of limestone powder in cement-based materials: mechanism, hydration and microstructures, *Constr. Build. Mater.* 181 (2018) 659–672, <https://doi.org/10.1016/j.conbuildmat.2018.06.075>.
- [72] K.J. Mun, S.Y. So, Y.S. Soh, The effect of slaked lime, anhydrous gypsum and limestone powder on properties of blast furnace slag cement mortar and concrete, *Constr. Build. Mater.* 21 (2007) 1576–1582, <https://doi.org/10.1016/j.conbuildmat.2005.09.010>.
- [73] G. Menéndez, V. Bonavetti, E.F. Irassar, Strength development of ternary blended cement with limestone filler and blast-furnace slag, *Cem. Concr. Compos.* 25 (2003) 61–67, [https://doi.org/10.1016/S0958-9465\(01\)00056-7](https://doi.org/10.1016/S0958-9465(01)00056-7).
- [74] T. Boubekeur, B. Boulekbache, K. Aoudjane, K. Ezziene, E.-H. Kadri, Prediction of the durability performance of ternary cement containing limestone powder and ground granulated blast furnace slag, *Constr. Build. Mater.* 209 (2019) 215–221, <https://doi.org/10.1016/j.conbuildmat.2019.03.120>.
- [75] C. Papohunda, B. Akinbile, A. Shittu, Structure and properties of mortar and concrete with rice husk ash as partial replacement of ordinary Portland cement – a review, *Int. J. Sustain. Built Environ.* 6 (2017) 675–692, <https://doi.org/10.1016/j.ijbsbe.2017.07.004>.
- [76] G. Arliguie, H. Hornain, *GranDuBé: Grandeurs associées à la durabilité des bétons*, Presses de l'École nationale des ponts et chaussées, 2007. <https://books.google.fr/books?id=0hUQMCJG9q0C>.